Assignment Report: Statistics and Trends
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1. Introduction

This report analyzes a dataset of 1,000 healthcare records to identify patterns between demographic factors (Age), physical attributes (BMI), and the prevalence of chronic diseases. By utilizing the four statistical moments and diverse visualization techniques, we aim to characterize the health profile of the study group and identify potential risk correlations.

2. Analysis of Statistical Moments

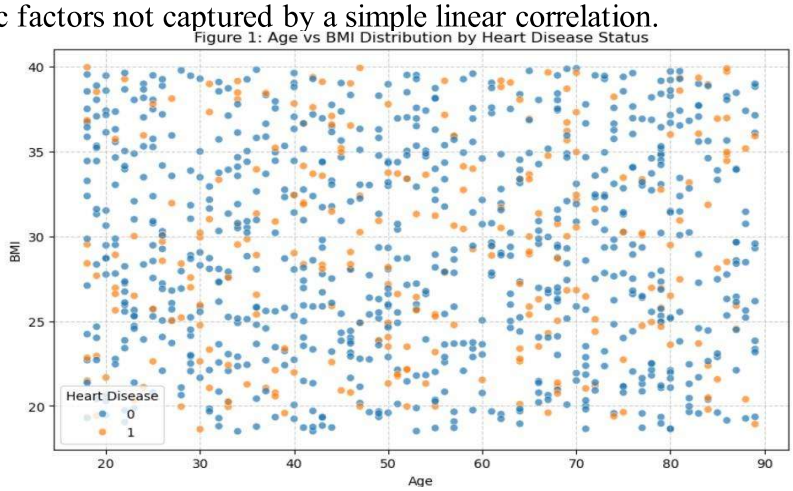
To understand the distribution of our primary metrics, we calculate the four main statistical moments for Age and BMI.

Moment	Age	BMI	Description
Mean	52.88	29.21	The average individual is middle-aged and overweight.
Variance	439.28	38.39	Age is widely spread; BMI is more concentrated.
Skewness	-0.01	0.06	Both are near 0, indicating a symmetrical distribution.
Kurtosis	-1.22	-1.20	Negative values indicate a "flat" (platykurtic) distribution.

The near-zero skewness suggests that the dataset is balanced across all age and weight groups. The negative kurtosis (-1.22 and -1.20) indicates a "platykurtic" shape, meaning the values are spread uniformly across a wide range rather than being bunched around a single peak. This ensures that the analysis is not biased toward any specific sub-group.

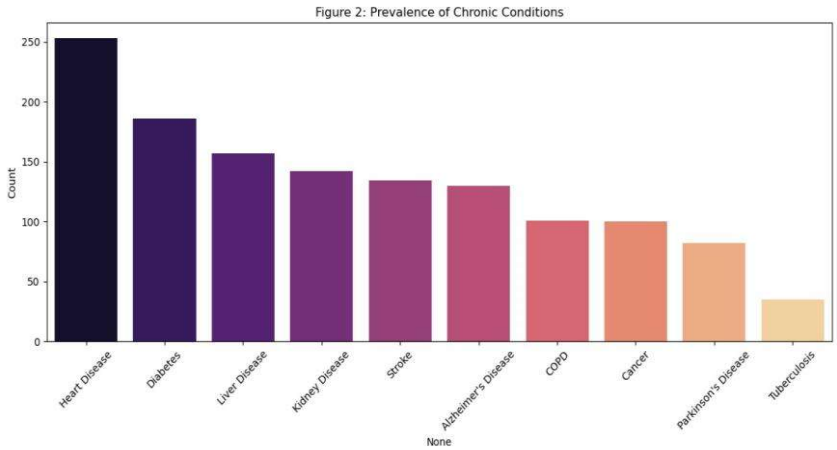
3. Relational Analysis (Age vs. BMI)

This figure demonstrates the relationship between Age and BMI. Contrary to common assumptions, there is no strong upward trend showing BMI increasing significantly with age in this cohort. The markers for heart disease (colored) appear sporadically across the entire scatter field. This suggests that while age and weight are risk factors, their impact is complex and likely interacts with other lifestyle or genetic factors not captured by a simple linear correlation.



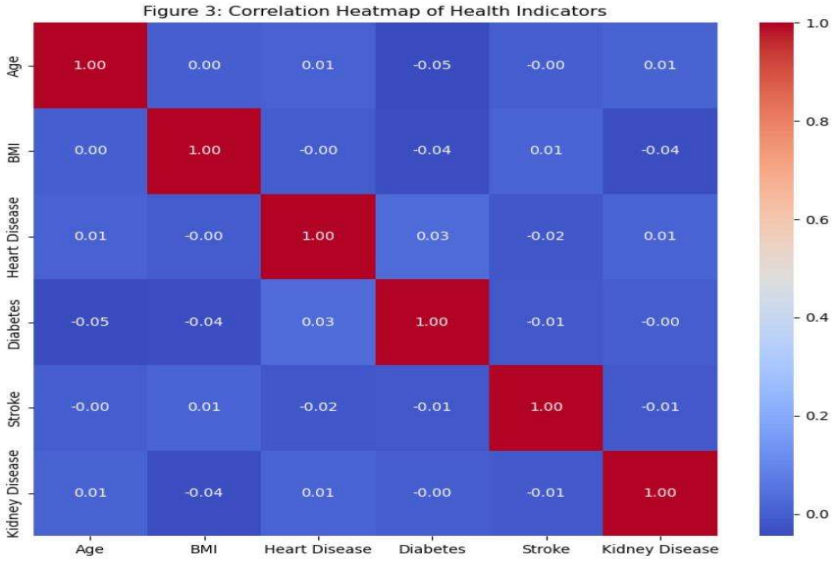
4. Categorical Analysis (Disease Prevalence)

This Figure is the Categorical Plot which shows the frequency of ten chronic conditions. Heart Disease and Diabetes are the most prevalent, which aligns with the high mean BMI of 29.21 (borderline obese). The prevalence of these "lifestyle diseases" suggests a population under metabolic stress. In contrast, neurological and infectious diseases like Alzheimer's and Tuberculosis show much lower frequency, highlighting the specific public health challenges facing this demographic.



5. Statistical Correlation Heatmap

This figure is the statistical Plot that examines the interdependencies between variables. The Pearson correlation coefficients remain low across the board (mostly below 0.10). This demonstrates that no single variable (like Age or BMI) is a standalone predictor for any targeted disease in this dataset. The independence of these variables tells that chronic diseases in this population are "multifactorial," requiring a combination of many different indicators for accurate prediction.



6. Conclusion

The population studied is characterized by a symmetrical, broadly spread demographic. The analysis shows that while chronic conditions like Diabetes are common, they are not strictly tied to a specific age or weight class. This highlights the importance of comprehensive screening rather than relying on individual physical metrics for health assessment.